B. Tech.(Computer Engg.) V Semester Examinations

Session 2024-25

Time: 1 hr. **Computer Networks (CEN - 505) Sessional-II** Max. Marks: 15

Note: Attempt All Questions

1. **CO3**	Explain the various framing techniques used at the Data Link Layer. A bit-stuffing framing method uses a flag sequence 0111110. If the data to be transmitted is 01111001111101110, calculate how many additional bits are transmitted due to bit stuffing.	**3**
2. **CO3**	In the Go-Back-N ARQ protocol, explain why the sender's window size is 2^n -1, while the receiver's window size is limited to 1. Additionally, describe how the protocol handles the following scenarios: 1. Duplicate frames 2. Lost frames 3. Lost acknowledgments	**4**
3. **CO3**	The values of parameters for the Stop-and-Wait ARQ protocol are as given below: • Bit rate of the transmission channel = 1 Mbps • Propagation delay from sender to receiver = 0.75 ms. • Time to process a frame = 0.25 ms. • Number of bytes in the information frame = 1980 while number of bytes in the acknowledgement frame = 20. Number of overhead bytes in the information frame = 20. Assume that there are no transmission errors. Calculate the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters.	**4**
4. **CO4**	In a Slotted ALOHA system, if the probability of collision for a frame is 0.5, calculate the efficiency and throughput of the system?	**4**

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